Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/27/2017
PI Name:	Wang, Kegang Ph.D.		
Project Title:	Testing Analytical and Numerical Models in Phase Coarsening using NASA Physical Sciences Informatics System		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kwang@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	321-674-8799
Organization Name:	Florida Institute of Technology		
PI Address 1:	Mechanical and Aerospace Engineering		
PI Address 2:	150 W University Blvd		
PI Web Page:			
City:	Melbourne	**State:**	FL
Zip Code:	32901-6982	**Congressional District:**	8
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	09/16/2016	**End Date:**	09/15/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Draper, Jeffrey	**Contact Phone:**	256.544.6680
Contact Email:	jeffrey.draper-1@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Glicksman, Martin Ph.D. (Florida Institute of Technology)		
Grant/Contract No.:	NNX16AT74G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A quantitative understanding of microstructure evolution by phase coarsening is crucial to the optimization of processing procedures for controlling the final structure and properties of multiphase materials. The volume fraction of the coarsening phase plays an integral role in the kinetics. The existing theoretical work has been limited to some volume fractions. The most ground-based experimental study of phase coarsening has been limited in model alloy systems. There were four microgravity experimental works for phase coarsening in Sn-Pb mixtures. The microgravity experiments greatly reduced the rate of sedimentation and allowed the dynamics of the phase coarsening to be investigated carefully and accurately. The specific thrusts of this project therefore lie in advancing the state of the art of understanding phase coarsening using NASA Physical Sciences Informatics (PSI) System. The proposed research project is to test existing analytical and numerical models for phase coarsening using data in		

	NASA Physical Sciences Informatics System. The primary goals of this project are: (1) We will test the diffusion screening theory of phase coarsening by virtue of the microgravity experiments in NASA Physical Sciences Informatics System; (2) We will test numerical modeling such as phase-field simulation; (3) Large-scale three-dimensional phase-field simulations will be conducted utilizing the microgravity experiments in NASA Physical Sciences Informatics System; (4) We will carry out a three-way comparison among theory, simulation, and microgravity experiments; (5) After the testing, an improved theory and a more accurate numerical modeling may be developed if needed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research is potentially improving our understanding of fundamental materials phenomena, and advancing the state of the art in both theory and experiment as it pertains to microstructural evolution in materials science. Our research will help engineers improve process designs in economically important areas, including liquid-phase sintering for the production of hard tool materials, and industrial crystallization of commodity chemicals and fertilizers, where controlling the particle size distribution underlies important safety and manufacturing issues. The improved knowledge of phase coarsening in this project will have significant impact on alloy design and service-life strategies for high-temperature materials. Our work has affected multiple disciplines such as materials science, chemistry, physics, and nanotechnology within the field of statistical mechanics modeling for pattern formation. This simulation is of significant interest for the physical sciences and technologies community.
Task Progress:	We have successfully developed a 2-D phase-field code along with a multi-particle diffusion code. We have conducted 2-D phase-field simulations for phase coarsening at a volume fraction of VV=0.8, and accomplished 3-D multiparticle diffusion calculations for phase coarsening at a volume fraction of VV=0.15. We compared these simulation results qualitatively with archived data derived from Coarsening in Solid-Liquid Mixtures (CSLM) experiments. Progress in these directions are included in a paper published in the JOM (Journal of Metals); see Bibliography section. We extracted concentration distributions from the initial microstructures measured at the earliest time archived from the CSLM experiments. To conduct quantitative comparison of simulations with archived experimental data from CSLM, NASA data were used as input for our phase-field code, allowing simulated evolution of coarsened microstructures for the Sn-Pb alloy mixtures processed aboard International Space Station (ISS). Finally, we shall compare the evolved microstructures from phase-field simulations with data recorded from the CSLM experiments for those volume fractions archived in NASA's PSI system, in order to test the numerical model critically and assess its reliability and limitations. We also plan to compare values of the mean and maximum particle sizes from the CSLM experiments with theoretical predictions from diffusion screening theory, in order to check the accuracy of this advanced theory of phase coarsening. We plan to compare the PSDs obtained from microgravity-processed alloys with our computational and theoretical predictions. Lastly, we plan to compare these experimentally derived coarsening rate constants with those based on our computational and theoretical predictions.
Bibliography Type:	Description: (Last Updated: 10/30/2021)
Abstracts for Journals and Proceedings	Wang KG, Yan H, Glicksman ME. "Large-scale phase-field simulations for microstructure evolution during phase coarsening." Invited talk in Modeling and Simulation, Frontiers in Materials Processing Applications, Research and Technology 2017, Bordeaux, France, July 9-13, 2017. Frontiers in Materials Processing Applications, Research and Technology 2017, Bordeaux, France, July 9-13, 2017. , Jul-2017
Abstracts for Journals and Proceedings	Glicksman ME. "Capillary Perturbations: Interface Fields Affecting Solidification." Plenary Lecture at 6th Decennial International Conference on Solidification Processing, Beaumont Estate, Old Windsor, UK, July 25-28, 2017. 6th Decennial International Conference on Solidification Processing, Beaumont Estate, Old Windsor, UK, July 25-28, 2017. , Jul-2017
Articles in Peer-reviewed Journals	Wang KG, Glicksman ME. "Testing numerical modeling of phase coarsening by microgravity experiments." Journal of The Minerals, Metals & Materials Society. Online August 30, 2017. https://doi.org/10.1007/s11837-017-2553-x , Aug-2017